

IOT-ASSIGNMENT

WASSMALL

Date _____
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Q1

I will choose ESP32 because it has built in Wifi/Bluetooth higher processing power, dual core CPU more ram and ADC channels. It can monitor soil moisture, control a relay for the pump, and upload data every 5 second without external modules. If internet is not required arduino UNO is sufficient for sensing and pump control because of its simplicity and low cost.

Q2

Ans use LoRa for long range sensor communication across 2 km and MQTT over the internet from a LoRa gateway to the cloud. Wifi has high band which but short range and high power consumption bluetooth for 500 nodes LoRa offers kilometers of coverage with very low power.

Q3

Ans Check sensor with another controller inspect wiring, verify supply voltage measure ADC reference compare raw ADC values and debug software scaling/filtering to isolate the fault.

Q4

Ans use deep sleep efficient voltage regulator, larger battery solar charging, transmit data less frequently ^{unused} optimize code disable ~~unused~~ peripheral and wake only on timer.

Q5

Ans We use CAN because it provide deterministic low latency reliable communication with error detection making it ideal for vehicle breaking

Q6

Ans Place the sensor near the center of the room at about 1.2-1.5m height away from windows. A.C. units and sunlight. Poor placement causes biased reading.

Q7

Ans Analog sensor excel when continuous high resolution signals low latency or low cost are needed

Q8

Ans Sensor $\rightarrow$ ESP32 $\rightarrow$ wifi $\rightarrow$ Router
 Mobile App $\leftarrow$ Database $\leftarrow$ Cloud

Polynomial fitting

Sensors

power

Controller

wifi router

cloud app

Solutions:-

Redundancy

watchdog

checksum

UPS

local storage

error handling

Q9

Ans.

Possible attacks: data theft, firmware modification, botnet enrolment, DoS, spoofing.
 Security: WPA3, TLS authentication, secure boot, Signed OTA, fire wall, strong passwords.

Q10

Ans.

Conflicts: GPIO shortage, Timing Conflicts, I2C sharing, relay noise, blocking code.
 Use freeRTOS, tasks, interrupts, pullups and proper pin planning.

Q11

Ans.

Buffer reading in SPIFFS/SD cards with time stamps. Retry upload periodically and delete only after cloud acknowledgement.

Q12

Ans.

Industrial components, watchdog, OTA, surge protection, conformal coating, wear leveling, robust error recovery, preventive diagnostics.

Q13

Ans.

Yes. Smart system automate decision using rules. Intelligent system use AI to learn and adapt.
 Example: Thermostat vs AI energy optimiser.

Q14

Ans.

Polling wastes CPU interrupts respond immediately to events to events and improves response for sensor buttons and communication.

Q15

Ans. Simple trade off: low cost sensor vs accuracy, wifi vs battery life, simple algorithm vs advanced analytics just depend on proper goals

Q16

Ans. Start $\rightarrow$ Read sensor $\rightarrow$ Wifi $\rightarrow$ Yes
Upload MQTT/HTTP $\rightarrow$ success? $\rightarrow$ Store if fail
No: Store locally $\rightarrow$ Retry periodically
 $\rightarrow$ Reconnect $\rightarrow$ Upload stored data $\rightarrow$ Delivered
after ACK $\rightarrow$ Repeat

Q17

Ans. Check update server $\rightarrow$ New firmware? $\rightarrow$
Download Verify $\rightarrow$ Check sum signature
Install $\rightarrow$ Reboot $\rightarrow$ Self test $\rightarrow$ success? if no
Continue if not rollback to previous
firmware and Report error.